

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA14

COURSE NAME: Compiler Design

COURSE OUTCOME COVERED

CO4: Examine intermediate code generation techniques to enhance code optimization (BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 4

Total Marks:40

Annexure A

CLASS TEST

Code of Conduct:

*I Ramya S S/192421226 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Ramya S S

Q. No	Understanding of Concepts (25)	Explanation (25)	Application (20)	Organization(15)	Accuracy(10)	Total Score (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	

5	/ 4	/ 4	/ 4	/ 4	/ 4	
OVERALL RUBRICS VALUE						
	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 25	/ 30	/ 20	/ 15	/ 10	/ 100

Annexure A

CLASS TEST

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	15%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps

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COMPILER DESIGN - CSA1407

SLOT - D

CO4AT1 CLASS TEST

① Construct three address code

if $((a < b) \text{ and } ((c < d) \text{ or } (a > d)))$ then $z = x + y * z$

else

 $z = z + 1$ Ans: Three address code

L1:

if $a < b$ goto L2

goto L5

L2:

if $c < d$ goto L4

goto L3

L3:

if $a > d$ goto L4

goto L5

L4:

 $t1 = y * z$ $t2 = x + t1$ $z = t2$

goto L6

L5:

 $t3 = z + 1$ $z = t3$

L6:

Conditions : $a < b$ $c < d$ $a > d$ Labels : L1, L2, L3, L4, L5

39/40

② Translate $(a < b)$ and $(c < d)$ or $(e < f)$ into three address statements using Back patching.

Ans: Backpatch statement $[(a < b) \text{ and } (c < d)] \text{ OR } (e < f)$

1. if $(a < b)$ goto 3
2. goto 5
3. if $(c < d)$ goto 7
4. goto 5
5. if $(e < f)$ goto 7
6. goto 8
7. True
8. False

③ Consider the arithmetic expression:

$$A = (C - A + B) * ((C - A + B) * C)$$

Ans: $t_1 = -A$
 $t_2 = t_1 + B$
 $t_3 = -A$
 $t_4 = t_3 + B$
 $t_5 = t_4 * C$
 $t_6 = t_2 * t_5$
 $A = t_6$

Quadruple:

Operator	arg1	arg2	Result
uminus	A	-	t1
+	t1	B	t2
uminus	A	-	t3
+	t3	B	t4
*	t4	C	t5
*	t2	t5	t6
=	t6	-	A

④ Construct three address code using the following production for Boolean expression $(a > b)$ or $(c < d)$ or $(e < f)$

$B \rightarrow B_1 || B_2$

$B \rightarrow B_1 \&\& B_2$

$B \rightarrow ! B_1$

$B \rightarrow E_1 \text{ relop } E_2$

$B \rightarrow \text{true}$

$B \rightarrow \text{false}$

Three address Code

Ans: L1: if $(a > b)$ goto LTRUE
goto L2

L2:

if $c < d$ goto LTRUE

goto L3

L3:

if $e < f$ goto LTRUE

goto LFALSE

LTRUE: // Boolean expression is true

goto LEND

LFALSE: '

// Boolean expression is false

LEND: